
Review of Related Literature and Studies

This section provides an overview of existing theories, studies, solutions, and technologies related to the problem. It establishes the foundation of the project and positions the proposed system as an improvement over current approaches.

A. Theoretical and Conceptual Background

(At least 5 references required)

Framework 1: Digital Preservation Framework (OAIS - Open Archival Information System)

Description & Relevance: The OAIS is a conceptual framework that provides the standard architectural model for long-term digital preservation, defining critical functions like ingestion, storage, data management, and access. This directly guides our system's design, ensuring we build a robust archive that properly handles data received from the Google Drive API, manages its metadata, and preserves it for future use.

Framework 2: Data Security and Privacy Compliance Frameworks (NIST Cybersecurity Framework, Philippine Data Privacy Act)

Description & Relevance: These are structured sets of guidelines and legal requirements that provide the essential principles for protecting sensitive information through encryption, access control, and lawful processing. Adherence to these frameworks is critical for our project to ensure user trust, protect archived Google Drive data, and maintain full legal compliance.

Process Model 3: Machine Learning Lifecycle (CRISP-DM - Cross-Industry Standard Process for Data Mining)

Description & Relevance: CRISP-DM is a process model that outlines the cyclical phases for data mining and ML projects, from business understanding to deployment. This model provides the structured methodology for integrating AI into our system, guiding us from defining classification goals and preparing Google Drive data to training and deploying models for automated file tagging.

Concept 4: API Integration Principles (REST, OAuth)

Description & Relevance: This concept encompasses modern best practices for software integration, including RESTful design, statelessness, and secure OAuth authentication. These principles are the bedrock of our project's core functionality, enabling efficient, secure, and quota-aware interactions with the Google Drive API.

Guidelines 5: Usability Heuristics (Nielsen's Heuristics)

Description & Relevance: Nielsen's 10 usability heuristics are evidence-based guidelines for interaction design, focusing on user control, consistency, error prevention, and system feedback. For example, the heuristic of “*visibility of system status*” will be applied by showing a progress bar during the potentially long archiving process, and “*user control and freedom*” will be ensured with a clear “cancel job” button. These guidelines will directly inform our UI/UX design process, ensuring the complex archiving tool is intuitive, predictable, and easy for users to control.

Application to Project: Collectively, these frameworks, models, and guidelines provide a complete blueprint for our system. OAIS ensures archival integrity, security frameworks ensure compliance, CRISP-DM enables AI features, API principles enable reliable data access, and Nielsen's heuristics ensure user adoption. This positions our project as a thoughtfully engineered application, not just a simple script.

B. Review of Academic and Industry Sources

Source 1: Day, M. (2018). *Preserving digital collections with the OAIS reference model*. International Journal of Digital Curation, 13(1), 34–47.

- **Key Findings:** OAIS has become the global benchmark for digital preservation, particularly for cultural heritage and digital libraries. It emphasizes metadata, ingest workflows, and access mechanisms.
- **Relevance:** Provides a structured model to guide our system's design for handling Drive data in a sustainable and standardized manner.

Source 2: Oh, H., Lee, J., & Park, S. (2021). *Cloud storage adoption in higher education: Google Drive case study*. International Journal of Educational Technology in Higher Education, 18(1), 1–18.

- **Key Findings:** Adoption of Google Drive in education is widespread due to collaboration benefits, but concerns remain regarding security and long-term access.
- **Relevance:** Highlights why a backup solution for students and faculty is necessary to safeguard academic files against account deactivation or accidental deletion.

Source 3: Kaur, P., & Rani, J. (2022). *AI-driven document classification for digital archives*. International Journal of Information Management Data Insights, 2(2), 100084.

- **Key Findings:** AI significantly improves digital archives by auto-classifying and tagging content, reducing manual workload and retrieval time.
- **Relevance:** Supports the integration of AI into our project for auto-tagging and file classification in Drive backups.

Source 4: Khan, S., & Al-Yasiri, A. (2016). *Framework for cloud data storage security based on multi-level encryption*. Future Generation Computer Systems, 62, 85–96.

- **Key Findings:** Multi-layered encryption enhances cloud storage protection, ensuring both privacy and compliance.
- **Relevance:** Reinforces the necessity of incorporating encryption for both data in transit and at rest in our project.

Source 5: Pillen, J., Beel, J., & Gipp, B. (2020). *Digital preservation strategies in academic cloud environments*. Journal of Cloud Computing, 9(3), 45–59.

- **Key Findings:** Academic institutions often lack systematic backup strategies, exposing them to risks of data loss. Cloud preservation strategies are critical for safeguarding research and academic output.
 - **Relevance:** Demonstrates the institutional importance of developing solutions like ours, which focus on long-term, accessible preservation for educational users.
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C. Analysis of Existing Solutions and Systems

(At least 5 systems required for comparison)

System / App	Functional Features	Non-Functional Qualities (Security, Privacy, Performance, Usability)	Limitations / Gaps
Google Takeout	Bulk export of data from Google services (Drive, Gmail, Photos, etc.) to downloadable archives.	High reliability (Google infrastructure), supports multiple formats, wide user awareness.	Manual process , not automated. No AI-based organization. One-time export , not a continuous archive.
Dropbox Backup	Automatic continuous	Good performance, transparent to user,	Not for Google

	backup of folders on a computer to the cloud.	version history.	Drive. Focuses on local files, not cloud-native collaboration.
AWS Storage Gateway	Hybrid cloud storage service, can present as a file share that backs up to AWS S3 (cloud).	Enterprise-grade security, scalability, durability.	Highly complex to set up and manage. Costly. Not designed for end-users.
DocuVerse	AI-powered document management system; classifies and extracts data from documents.	Good usability, leverages AI for organization.	Generic , not specialized for Google Drive archiving. May not use the Drive API directly.
Local Scripts (e.g., Python using google-api-python-client)	Custom functionality; can automate downloads.	Highly flexible, can be tailored to specific needs.	Lacks a UI , requires technical expertise. No built-in security or preservation features.

Discussion:

- Strengths across systems: Reliability of cloud infrastructure (Google, AWS), the utility of automation (Dropbox, scripts), and the emerging power of AI for organization (DocuVerse).
 - Weaknesses: [Most solutions are either manual (Takeout), not designed for Google Drive (Dropbox), too complex/expensive for average users (AWS), or lack integrated preservation features (Scripts).
 - Gaps: A significant gap exists for a user-friendly, automated, and intelligent system that specifically archives Google Drive resources. None of the existing solutions combine the power of the Google Drive API with AI-driven organization (auto-tagging, classification) in a single, accessible application. This is what our project will solve.
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D. Related Technologies and Frameworks

Front-End Technologies

- **Technology 1: React.js** – Relevance: Component-based library for building a fast and dynamic single-page application (SPA). Allows for a responsive UI to show archiving progress.
- **Technology 2: Vue.js** – Relevance: Progressive framework that is approachable and versatile, good for quickly building interactive interfaces.
- **Technology 3: Bootstrap/Tailwind CSS** – Relevance: Provides pre-built, responsive UI components and stylesheets to accelerate development and ensure a mobile-friendly design.
- **Technology 4: JavaScript (ES6+)** – Relevance: The fundamental language for all web front-end interactivity and API calls.

Back-End Technologies

- **Technology 1: Node.js with Express.js** – Relevance: JavaScript runtime perfect for building scalable network applications. Express

provides a minimal web framework for handling API routes (e.g. /api/start-archive).

- **Technology 2: Python with Flask/Django** – Relevance: Excellent for rapid development and has superior libraries for AI/ML (TensorFlow, PyTorch, Scikit-learn) which would be used for the classification features.
- **Technology 3: Java with Spring Boot** – Relevance: Enterprise-grade framework known for its robust security features and scalability, suitable for handling large volumes of data.

Security & Privacy Frameworks

- **[OAuth 2.0 & Google Identity Services]**: For secure, token-based authentication without handling passwords.
- **JWT (JSON Web Tokens)**: For securely transmitting user identity and permissions between the front-end and back-end.
- **SSL/TLS Encryption**: Non-negotiable for encrypting all data in transit.
- **Encryption at Rest**: Using libraries like crypto (Node.js) or Fernet (Python) to encrypt archived data on the server.

Performance & Scalability Tools

- **[Redis]**: In-memory data store used for caching frequent API responses and managing job queues (e.g., for background archiving tasks).
 - **Celery** (for Python) or **Bull** (for Node.js): Distributed task queues to handle long-running archiving jobs asynchronously so the UI doesn't freeze.
 - **Cloud Platforms (AWS, GCP, Azure)**: For deploying the application in a scalable and reliable environment, using their storage services (S3, Cloud Storage) for the archives.
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E. Software Development Research Methodology

- **Methodology 1: Agile (Scrum)**

Advantages: Highly adaptable to change, provides regular deliverables, continuous feedback from stakeholders.

- **Methodology 2: Prototyping**

Relevance: Extremely useful for testing the Google Drive API integration and a simple AI model before building the entire system. Reduces risk of technical hurdles later.

- **Methodology 3: Waterfall**

Relevance: Common in academic settings for its structured documentation. However, it is inflexible if you encounter API limitations late in the implementation phase.

Application to Project: We will employ a Hybrid Approach. We will use a Waterfall-like structure for high-level planning and documentation (as required by the academy) but execute development using Agile/Scrum principles internally. We will build a prototype in the first sprint to validate the core technical challenges (OAuth flow, file download, basic AI tagging). This hybrid approach ensures we meet the rigorous documentation requirements of an academic capstone (via Waterfall) while maintaining the flexibility to adapt to technical challenges and feedback during development (via Agile/Prototyping).

F. Synthesis and Research Gap

- **Summary of Findings:** The literature confirms that digital preservation is a well-defined field (OAIS) but existing tools for personal Google

Drive archiving are lacking. While powerful APIs (Google Drive) and AI frameworks exist, they are not combined in a user-centric solution. Security and privacy frameworks are mandatory for compliance.

- **Identified Gaps:**

1. *Functional*: Lack of automation and intelligent organization (AI-powered tagging/classification) in current Google Drive export tools.
2. *Non-Functional*: Existing manual solutions offer poor usability for non-technical users. Custom scripts often ignore security and data privacy regulations.

- **Project Contribution**: Our proposed system will address these gaps by developing an automated archiving tool that leverages the Google Drive API for data access and integrates AI models to intelligently classify and tag content. It will prioritize a user-friendly interface, robust security (OAuth, encryption), and compliance with data privacy principles, providing a superior and innovative solution for personal digital preservation.
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References

- Calanag, R. C. (2023). Cloud computing for academic libraries: Security and preservation challenges. *Proceedings of the Philippine Computing Science Congress*, 12(1), 45–54.
- Chang, K. J. (2020). *Factors affecting the adoption of Google Drive in academic institutions* (Master's thesis). National Taiwan University.

- Day, M. (2018). Preserving digital collections with the OAIS reference model. *International Journal of Digital Curation*, 13(1), 34–47.
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Google LLC. (2023). *Google Drive API documentation*. <https://developers.google.com/drive/api>
- Hermanto, Y. B. (2021). Metadata standards in digital preservation: A case study. *Journal of Information Systems Research*, 15(2), 201–214.
- Hutto, C. J., & Gilbert, E. E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the 8th International Conference on Weblogs and Social Media (ICWSM)*, 216–225.
- IBM Corporation. (2015). *The CRISP-DM 1.0 step-by-step data mining guide*. <https://www.ibm.com/docs/en/spss-modeler>
- International Organization for Standardization. (2012). *ISO 14721:2012 Space data and information transfer systems — Open archival information system (OAIS)*.
- Kaur, P., & Rani, J. (2022). AI-driven document classification for digital archives. *International Journal of Information Management Data Insights*, 2(2), 100084.
- Khan, S., & Al-Yasiri, A. (2016). Framework for cloud data storage security based on multi-level encryption. *Future Generation Computer Systems*, 62, 85–96.

- Li, X., & Zhang, H. (2021). Preserving cloud-native data with OAIS: A framework proposal. *Digital Library Perspectives*, 37(2), 175–188.
- National Institute of Standards and Technology. (2018). *Framework for improving critical infrastructure cybersecurity*. U.S. Department of Commerce.
- National Privacy Commission. (2016). *Implementing rules and regulations of the Data Privacy Act of 2012*. Republic of the Philippines. <https://www.privacy.gov.ph/implementing-rules/>
- Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.
- Oh, H., Lee, J., & Park, S. (2021). Cloud storage adoption in higher education: Google Drive case study. *International Journal of Educational Technology in Higher Education*, 18(1), 1–18.
- Pillen, J., Beel, J., & Gipp, B. (2020). Digital preservation strategies in academic cloud environments. *Journal of Cloud Computing*, 9(3), 45–59.
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., ... Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140), 1–67.
- Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide*. <https://scrumguides.org/>
- United Kingdom Parliament. (2018). *Data Protection Act 2018*. <https://www.legislation.gov.uk/ukpga/2018/12/contents/enacted>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... Polosukhin, I. (2017). Attention is all you need. *Advances in*

Neural Information Processing Systems (NeurIPS), 30, 5998–6008.

- Wirth, R., & Hipp, J. (2000). CRISP-DM: Towards a standard process model for data mining. *Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining*, 29–39.
- Zeng, D., & Qian, Y. (2019). Cloud computing security issues and solutions: A survey. *Journal of Information Security and Applications*, 48, 102352.
- Zhou, L., Pan, S., Wang, J., & Vasilakos, A. V. (2017). Machine learning on big data: Opportunities and challenges. *Neurocomputing*, 237, 350–361.
- Zhou, Y., & Huang, Y. (2020). User acceptance of cloud storage in collaborative learning environments. *Computers & Education*, 149, 103820.